Chapter 11

Gauge Invariance

Road map

Gauge theory distinguishes physical fields from redundant potentials used to describe them. Maxwell theory is the paradigmatic example. We first review Minkowski notation and express Maxwell's equations by the antisymmetric field tensor. Differential forms explain locally why a four-potential exists. The potential displays gauge freedom, permits Lorenz gauge fixing, and leads to a compact action principle. We finish by deriving the field equations from that action and explaining how global and local symmetries enter Noether's first and second theorems.

11.1 The language of gauge theory

The following terms must be kept conceptually distinct.

Gauge freedom
A description contains redundant degrees of freedom: more than one mathematical field represents the same physical state.

Gauge transformation
A transformation moves from one representative to another without changing the physical observables.

Gauge invariance
Observable fields and the physical predictions are unchanged under gauge transformations.

Gauge fixing One chooses a representative from every equivalence class by imposing an additional condition.

Gauge group The set of all gauge transformations, with composition as its group operation.

Gauge field A connection-like field introduced so that an interacting theory can respect a local symmetry.

Maxwell electromagnetism makes all of these ideas explicit and is most transparent in relativistic Lagrangian form.

11.2 Minkowski spacetime and index notation

Use spacetime coordinates

$$x^\mu = (ct, x^1, x^2, x^3) = (ct, x, y, z), \qquad \mu = 0, 1, 2, 3.$$

The underlying spacetime is denoted M, and in standard inertial coordinates it is $\mathbb{R}^4$. With signature $(+, -, -, -)$, the metric is

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

It lowers and raises indices:

$$x_\mu = \eta_{\mu\nu} x^\nu, \qquad x^\mu = \eta^{\mu\nu} x_\nu.$$

Therefore

$$x_\mu = (ct, -x, -y, -z).$$

From now on, a repeated upper–lower index is summed from 0 to 3; this is the Einstein summation convention.

The covariant derivative components are

$$\partial_\mu := \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right),$$

and

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu.$$

Under a Lorentz transformation $x \mapsto x'$,

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu.$$

The metric is the structure preserved by Lorentz transformations:

$$\Lambda^T \eta \Lambda = \eta.$$

11.3 Maxwell theory in tensor form

11.3.1 Charge and current density

Combine the charge density $\rho = \rho(t, \boldsymbol{x})$ and current density $\boldsymbol{j} = \boldsymbol{j}(t, \boldsymbol{x})$ into the four-current

$$j^\mu = (c\rho, \boldsymbol{j}).$$

This field is the source of the electromagnetic field.

11.3.2 The electromagnetic field tensor

The electric and magnetic fields are assembled into the antisymmetric tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}, \qquad F^{\mu\nu} = -F^{\nu\mu}.$$

Lowering both indices gives

$$F_{\mu\nu} = \eta_{\mu\alpha}\eta_{\nu\beta}F^{\alpha\beta} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}.$$

The time–space entries change sign when both indices are lowered, whereas the purely spatial entries do not.

11.3.3 The inhomogeneous equations

Gauss's law and the Ampère–Maxwell law combine into

$$\boxed{\partial_\mu F^{\mu\nu} = \mu_0 j^\nu.} \tag{11.1}$$

For $\nu = 0$, this is

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}.$$

For the three spatial values of ν, it is

$$\nabla \times \boldsymbol{B} - \varepsilon_0\mu_0 \frac{\partial \boldsymbol{E}}{\partial t} = \mu_0 \boldsymbol{j}.$$

Here

$$c^{-2} = \varepsilon_0\mu_0.$$

11.3.4 The homogeneous equations

The absence of magnetic charge and Faraday's law combine into the cyclic identity

$$\boxed{\partial^\lambda F^{\mu\nu} + \partial^\mu F^{\nu\lambda} + \partial^\nu F^{\lambda\mu} = 0.} \tag{11.2}$$

In three-vector notation these equations are

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \nabla \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = 0.$$

11.3.5 Charge conservation

Apply ∂_ν to (11.1):

$$\mu_0 \partial_\nu j^\nu = \partial_\nu \partial_\mu F^{\mu\nu}.$$

The derivatives commute, so $\partial_\nu \partial_\mu$ is symmetric under $\mu \leftrightarrow \nu$, while $F^{\mu\nu}$ is antisymmetric. Explicitly,

$$\begin{aligned}\partial_\nu \partial_\mu F^{\mu\nu} &= \frac{1}{2}\partial_\nu \partial_\mu \big(F^{\mu\nu} + F^{\nu\mu}\big)\\ &= 0.\end{aligned}$$

Therefore

$$\boxed{\partial_\mu j^\mu = 0.} \tag{11.3}$$

In ordinary notation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \boldsymbol{j} = 0,$$

the continuity equation for electric charge.

11.4 Differential forms and the four-potential

11.4.1 The field strength as a two-form

Regard the field tensor as the differential two-form

$$F = \frac{1}{2} F_{\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

For an ordinary scalar function f,

$$\mathrm{d}f = (\partial_\lambda f)\mathrm{d}x^\lambda,$$

and the exterior derivative obeys the graded Leibniz rule

$$\mathrm{d}(f\omega) = \mathrm{d}f \wedge \omega + f\, \mathrm{d}\omega.$$

Because coordinate one-forms are exact,

$$\mathrm{d}(\mathrm{d}x^\mu) = 0.$$

Hence

$$\begin{aligned}\mathrm{d}F &= \frac{1}{2}(\partial_\lambda F_{\mu\nu})\mathrm{d}x^\lambda \wedge \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu\\ &= \frac{1}{3!}\big(\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu}\big)\mathrm{d}x^\lambda \wedge \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.\end{aligned}$$

The second line follows by antisymmetrizing the coefficient, because the wedge product is completely antisymmetric. Thus the homogeneous Maxwell equations are exactly

$$\boxed{\mathrm{d}F = 0.}$$

11.4.2 Poincaré's lemma

Poincaré's lemma states that on a contractible coordinate neighbourhood, every closed form is exact. Therefore $\mathrm{d}F = 0$ implies locally that

$$F = \mathrm{d}A$$

for a one-form

$$A = A_\mu \mathrm{d}x^\mu.$$

Now

$$\begin{aligned}\mathrm{d}A &= (\partial_\nu A_\mu)\mathrm{d}x^\nu \wedge \mathrm{d}x^\mu \\ &= \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.\end{aligned}$$

Consequently,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \tag{11.4}$$

The vector field $A^\mu(x)$ is the electromagnetic four-potential.

Substituting (11.4) into the cyclic expression in (11.2) gives six second-derivative terms that cancel pairwise, since partial derivatives commute. Equivalently,

$$\mathrm{d}F = \mathrm{d}^2 A = 0.$$

Thus the homogeneous Maxwell equations hold automatically once $F = \mathrm{d}A$.

The three-dimensional analogy

For a conservative force,

$$\nabla \times \boldsymbol{F} = 0$$

implies locally that $\boldsymbol{F} = -\nabla V$. Writing the force in terms of the scalar potential automatically enforces the curl equation. The four-dimensional statement $F = \mathrm{d}A$ plays the same role for electromagnetism.

11.5 The electromagnetic action

Let

$$S[A] = \int \mathcal{L}\,\mathrm{d}^4x$$

with Lagrangian density

$$\mathcal{L} = -\frac{1}{4\mu_0}F^{\mu\nu}F_{\mu\nu} - j_\mu A^\mu. \tag{11.5}$$

It is local: at a spacetime point it depends only on $A^\mu(x)$, its first derivatives, and the prescribed source. The classical field equation follows from

$$\delta S = 0$$

for variations that vanish at the boundary.

The Euler–Lagrange equation for A^ν is

$$\frac{\partial \mathcal{L}}{\partial A^\nu} - \partial^\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial^\mu A^\nu)} \right] = 0.$$

The source term gives

$$\frac{\partial \mathcal{L}}{\partial A^\nu} = -j_\nu.$$

To differentiate the kinetic term, first note

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha.$$

Varying it gives

$$\delta F_{\alpha\beta} = \partial_\alpha \delta A_\beta - \partial_\beta \delta A_\alpha.$$

Therefore

$$\begin{aligned}
\delta\left(-\frac{1}{4\mu_0} F^{\alpha\beta} F_{\alpha\beta}\right) &= -\frac{1}{2\mu_0} F^{\alpha\beta} \delta F_{\alpha\beta} \\
&= -\frac{1}{2\mu_0} F^{\alpha\beta} (\partial_\alpha \delta A_\beta - \partial_\beta \delta A_\alpha) \\
&= -\frac{1}{\mu_0} F^{\alpha\beta} \partial_\alpha \delta A_\beta.
\end{aligned}$$

In the last step, interchange α and β in the second term and use $F^{\beta\alpha} = -F^{\alpha\beta}$. Thus

$$\frac{\partial \mathcal{L}}{\partial(\partial^\mu A^\nu)} = -\frac{1}{\mu_0} F_{\mu\nu}.$$

The Euler–Lagrange equation becomes

$$-j_\nu + \frac{1}{\mu_0} \partial^\mu F_{\mu\nu} = 0,$$

or, after raising the free index,

$$\boxed{\partial_\mu F^{\mu\nu} = \mu_0 j^\nu.}$$

This reproduces the inhomogeneous Maxwell equations.

The action formulation is valuable because symmetries and conserved quantities can be analyzed systematically, and it provides the natural starting point for canonical quantization.

11.6 Gauge transformations and gauge freedom

Definition 11.1: Electromagnetic gauge transformation

For any smooth real scalar function $\Lambda = \Lambda(x)$, define

$$A^\mu(x) \longmapsto A'^\mu(x) = A^\mu(x) + \partial^\mu \Lambda(x).$$

The field strength is unchanged:

$$\begin{aligned} F'^{\mu\nu} &= \partial^\mu A'^\nu - \partial^\nu A'^\mu \\ &= \partial^\mu A^\nu - \partial^\nu A^\mu + \partial^\mu\partial^\nu\Lambda - \partial^\nu\partial^\mu\Lambda \\ &= F^{\mu\nu}. \end{aligned}$$

Thus the observable electric and magnetic fields are gauge invariant. The potential is not unique: all potentials related by such a transformation describe the same electromagnetic field.

In the source-free theory, the Lagrangian density depends only on F and is exactly invariant. With a prescribed source,

$$\begin{aligned} -j_\mu A'^\mu &= -j_\mu A^\mu - j_\mu\partial^\mu\Lambda \\ &= -j_\mu A^\mu - \partial^\mu(j_\mu\Lambda) + \Lambda\,\partial^\mu j_\mu. \end{aligned}$$

If the source is conserved, the change is a total derivative and therefore does not alter the action under the usual boundary conditions.

11.6.1 The gauge group

For standard Abelian Maxwell theory, the gauge parameters form the additive group

$$\mathcal{G} = C^\infty(M, \mathbb{R}) = \{\Lambda : M \to \mathbb{R} : \Lambda \text{ smooth}\},$$

with

$$(\Lambda_1 + \Lambda_2)(x) = \Lambda_1(x) + \Lambda_2(x).$$

Composing the transformations associated with Λ_1 and Λ_2 produces the transformation associated with $\Lambda_1 + \Lambda_2$. For this reason Maxwell theory is an Abelian gauge theory.

Why the potential is called a gauge field

When charged matter is included, promoting a constant phase symmetry of the matter field to a spacetime-dependent symmetry requires the electromagnetic potential and a covariant derivative. The transformation of A_μ compensates for derivatives of the local phase. In this sense the gauge field makes a local symmetry possible in the interacting theory.

11.7 Lorenz gauge and the wave equation

Definition 11.2: Lorenz gauge

The Lorenz gauge condition is

$$\boxed{\partial_\mu A^\mu = 0.}$$

Using (11.4),

$$\begin{aligned} \mu_0 j^\nu &= \partial_\mu F^{\mu\nu} \\ &= \partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= \partial_\mu\partial^\mu A^\nu - \partial^\nu(\partial_\mu A^\mu). \end{aligned}$$

Define the d'Alembert operator

$$\Box := \partial_\mu \partial^\mu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2.$$

Under the Lorenz condition, the second term vanishes and Maxwell's equations reduce to four decoupled inhomogeneous wave equations:

$$\boxed{\Box A^\nu = \mu_0 j^\nu.} \tag{11.6}$$

11.7.1 Why Lorenz gauge can be imposed

Start with an arbitrary potential A^μ and seek

$$A'^\mu = A^\mu + \partial^\mu \Lambda$$

satisfying $\partial_\mu A'^\mu = 0$. The condition is

$$\partial_\mu A^\mu + \partial_\mu \partial^\mu \Lambda = 0,$$

or

$$\Box \Lambda = -\partial_\mu A^\mu. \tag{11.7}$$

This is an inhomogeneous wave equation for Λ, and locally it can be solved for suitable initial and boundary data. With such a solution, A'^μ represents the same $F^{\mu\nu}$ and satisfies Lorenz gauge.

Residual gauge freedom

Lorenz gauge does not always select a unique representative. If Λ_0 satisfies $\Box\Lambda_0 = 0$, then

$$A^\mu \longmapsto A^\mu + \partial^\mu \Lambda_0$$

preserves $\partial_\mu A^\mu = 0$. A complete gauge fixing may require additional boundary or initial conditions.

11.8 A nonrelativistic analogy

For a particle in a conservative force field,

$$m\ddot{\boldsymbol{r}} = \boldsymbol{F}(\boldsymbol{r}), \qquad \nabla \times \boldsymbol{F} = 0.$$

Locally there is a scalar potential V with

$$\boldsymbol{F} = -\nabla V,$$

so the equation of motion is

$$m\ddot{\boldsymbol{r}} = -\nabla V.$$

The potential is not unique:

$$V \longmapsto V + \Delta V, \qquad \Delta V \in \mathbb{R} \text{ constant},$$

leaves $\boldsymbol{F}$ unchanged. The additive group $(\mathbb{R}, +)$ is the gauge group of this elementary example. A condition such as

$$V(\boldsymbol{r}_0) = 0$$

fixes the additive freedom. This simple model separates the same four ideas: nonunique potential, gauge transformation, invariant force, and gauge fixing.

11.9 Noether's first theorem

Let fields $\psi_i(x)$ have a Lagrangian density

$$\mathcal{L} = \mathcal{L}(\psi_i, \partial_\mu \psi_i, x).$$

Consider an infinitesimal transformation

$$\psi_i(x) \longmapsto \psi_i(x) + \delta\psi_i(x), \qquad \delta\psi_i = \varepsilon F_i(\psi_1, \psi_2, \ldots),$$

where ε is an infinitesimal constant parameter. The first variation is

$$\delta\mathcal{L} = \sum_i \frac{\partial \mathcal{L}}{\partial \psi_i}\delta\psi_i + \sum_{i,\mu} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\delta(\partial_\mu \psi_i).$$

Because variation and differentiation commute,

$$\delta(\partial_\mu \psi_i) = \partial_\mu \delta\psi_i.$$

Apply the product rule:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\partial_\mu \delta\psi_i = \partial_\mu &\left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\delta\psi_i\right] \\ &- \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\right]\delta\psi_i.\end{aligned}$$

Therefore

$$\begin{aligned}\delta\mathcal{L} = \sum_i &\left[\frac{\partial \mathcal{L}}{\partial \psi_i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\right)\right]\delta\psi_i \\ &+ \partial_\mu \left[\sum_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\delta\psi_i\right].\end{aligned} \tag{11.8}$$

On solutions of the Euler–Lagrange equations, the first line vanishes. If the transformation leaves the Lagrangian invariant, $\delta\mathcal{L} = 0$, then

$$\partial_\mu J^\mu = 0,$$

where

$$J^\mu = \sum_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}\delta\psi_i \tag{11.9}$$

is the Noether current.

Theorem 11.1: Noether's first theorem, basic field form

Every differentiable one-parameter global symmetry of the action gives, on solutions of the field equations, a conserved current

$$\partial_\mu J^\mu = 0.$$

11.10 Local gauge symmetry and Noether's second theorem

11.10.1 Why a constant gauge parameter is trivial for A_μ

If $\delta\Lambda$ is constant, then

$$\delta A_\mu = \partial_\mu \delta\Lambda = 0.$$

Thus the "global" part of the gauge transformation acts trivially on the electromagnetic potential alone, and the current obtained from (11.9) is zero. Electric charge conservation is already implied by the Maxwell equation and, in a full matter theory, is associated with the global phase symmetry of the charged matter field.

11.10.2 Arbitrary local parameters produce an identity

For the source-free electromagnetic field, take

$$\delta A_\nu = \partial_\nu \Lambda(x)$$

with arbitrary $\Lambda(x)$. The action is invariant because $\delta F_{\mu\nu} = 0$. The Euler–Lagrange expression is

$$E^\nu := \frac{1}{\mu_0}\partial_\mu F^{\mu\nu}.$$

Gauge invariance of the action gives

$$\begin{aligned} 0 = \delta S &= \int E^\nu \partial_\nu \Lambda \, \mathrm{d}^4 x \\ &= -\int (\partial_\nu E^\nu)\Lambda \, \mathrm{d}^4 x, \end{aligned}$$

after discarding a boundary term. Since $\Lambda(x)$ is arbitrary,

$$\partial_\nu E^\nu \equiv 0.$$

Indeed,

$$\partial_\nu E^\nu = \frac{1}{\mu_0}\partial_\nu \partial_\mu F^{\mu\nu} \equiv 0$$

by the symmetric–antisymmetric contraction. This is not a new equation of motion; it is a differential identity among the field equations.

Theorem 11.2: Noether's second theorem, informal form

If an action is invariant under transformations depending on arbitrary functions of spacetime, then its Euler–Lagrange equations satisfy corresponding differential identities. Local gauge symmetry therefore signals redundancy among the field equations rather than an independent ordinary conserved charge for every choice of the function.

With an external current, the Euler–Lagrange expression is

$$E^\nu = \frac{1}{\mu_0}\partial_\mu F^{\mu\nu} - j^\nu.$$

The same identity becomes

$$\partial_\nu E^\nu = -\partial_\nu j^\nu.$$

Thus gauge invariance of the sourced action requires the current conservation law $\partial_\nu j^\nu = 0$, consistently with (11.3).

What to retain from this chapter

1. Maxwell's equations are

$$\partial_\mu F^{\mu\nu} = \mu_0 j^\nu, \qquad \partial^\lambda F^{\mu\nu} + \partial^\mu F^{\nu\lambda} + \partial^\nu F^{\lambda\mu} = 0.$$

2. The homogeneous equations say $\mathrm{d}F = 0$; locally Poincaré's lemma gives $F = \mathrm{d}A$, or $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.
3. The transformation $A^\mu \mapsto A^\mu + \partial^\mu \Lambda$ leaves $F^{\mu\nu}$ unchanged. This is the local Abelian gauge freedom.
4. Lorenz gauge $\partial_\mu A^\mu = 0$ reduces the sourced equations to $\Box A^\nu = \mu_0 j^\nu$, and it can be reached by solving a wave equation for the gauge parameter.
5. Varying $\mathcal{L} = -F^{\mu\nu}F_{\mu\nu}/(4\mu_0) - j_\mu A^\mu$ reproduces the inhomogeneous Maxwell equations.
6. Noether's first theorem relates global one-parameter symmetries to conserved currents. Local gauge symmetry instead yields differential identities among Euler–Lagrange equations, the content of Noether's second theorem.
7. Charge conservation follows from $\partial_\nu\partial_\mu F^{\mu\nu} = 0$, and it is also the consistency condition for coupling a source to a gauge-invariant electromagnetic action.